Review Report for "Multi-Armed Bandits with Arriving Arms: Sequential Screening, Dynamic Regret, and Sublinear Guarantees"

Summary

This paper studies a stochastic multi-armed bandit problem in which new arms arrive over time and remain available, with regret measured relative to the best arm currently available. The problem is related to the ballooning multi-armed bandit setting studied by Ghalme et al. (2021), but the proposed UCB-AA algorithm takes a different approach. Specifically, it adopts a round-based elimination procedure in which newly admitted arms first undergo preliminary screening against incumbent arms, and the surviving arms then enter a formal elimination stage together with the incumbents. The authors provide regret bounds, an online extension, lower bounds for representative regimes, and simulation studies comparing regret, wasted pulls, and the size of the active set.

The idea of screening newly arrived arms before they enter full competition is intuitive, and the paper develops its theoretical analysis in a logically structured manner. However, I have several concerns about the regret decomposition, the relationship between the round schedule and the actual pull budget, and several steps in the proofs. At the same time, some details of the simulation settings and implementation are not sufficiently clear, which makes the numerical results difficult to reproduce and interpret. These issues may affect both the justification of the main regret guarantees and the assessment of the practical benefits of the proposed method.

The comments are based on my reading of the manuscript and my own implementation of the algorithm. Page and line references below refer to the page numbers at the top of each page and the line numbers in the left margin.

Major Comments

1. The regret decomposition and its interpretation need further clarification.

   (a) *Regret from incorrect elimination in earlier rounds.* In Section 3.2, the authors define $R_{T_0} = R_M + D_M$, where R_M is the roundwise regret measured relative to the best arm in the active set B_m. The paper then interprets $D_M = R_{T_0} - R_M$ as the regret caused by delays in admitting arms that arrive within a round. However, $A(m)$ contains all arms admitted up to round m, including arms that may have been eliminated previously. Under these definitions,

$$\begin{aligned} R_{T_0} &= \sum_{m=0}^{M} \sum_{t \text{ in round } m} (r^*(t) - r_{I_t}) \\ &= \underbrace{\sum_{m=0}^{M} \sum_{i \in B_m} N_{i,m}^{\pi} \Delta_i(m)}_{R_M} + \underbrace{\sum_{m=0}^{M} \sum_{t \text{ in round } m} \left(\max_{i \in \mathcal{A}_t} r_i - \max_{i \in A(m)} r_i \right)}_{D_M^{\text{arrival}}} \\ &\quad + \underbrace{\sum_{m=0}^{M} L_m \left(\max_{i \in A(m)} r_i - \max_{i \in B_m} r_i \right)}_{\text{loss from earlier elimination}}. \end{aligned} \tag{R.1}$$

Thus, under the definitions in the manuscript, $D_M = R_{T_0} - R_M$ contains both the second and the third terms in Equation (R.1), rather than only the regret due to delayed admission of new arms. If $\beta_{m,j}$ denotes an actual increase in the best available mean, the second term equals $\sum_m \sum_j (L_m - \tau_{m,j} + 1)\beta_{m,j}$, which is the contribution represented by Equation (36) (p. 56). The third term can remain positive after a better arm has been eliminated, even if no further arms arrive. Also, the sum of arrival increments need not equal the change in $\max_{i \in B_m} r_i$ used to define δ_m (p. 14, Lines 48–54). The authors should therefore consider how this issue affects the subsequent theoretical analysis and the stated regret guarantees.

(b) *Regret components in the simulations.* Reporting the contributions of the three terms in Equation (R.1) separately in the simulation study would help distinguish regret arising from learning within rounds, delayed admission of new arms, and earlier elimination. This might help assess whether the third term is negligible in the simulation settings and relate the empirical results more directly to the theoretical decomposition.

2. The sampling targets grow on an exponential scale with the round index m. With $\widetilde{\Delta}_m = 2^{-m}$, Algorithm 1 gives $n_m = \lceil 2 \cdot 4^m \log(N4^{-m}) \rceil$. Such increasing sampling requirements are standard in elimination-based bandit methods and are not a problem in themselves. In this paper, however, new arms continue to arrive while UCB-AA operates through increasingly long rounds, which raises the following concerns.

(a) *Only a few rounds may be reached with a budget of tens of thousands of pulls.* It seems that completing rounds $0, \ldots, M$ in UCB-AA may require far more than tens of thousands of pulls, even for modest values of M. For example, the paper uses $T = 60000$ in its simulations. In my implementation of the supplement's Scenario 2 with Bernoulli rewards and $N = e4^M$, completing all the prescribed rounds gave the following mean total pull counts $\overline{T}_0$ over 1000 repetitions per setting:

M	N	n_M	$\overline{T}_0$
5	2.78×10^3	2048	8.64×10^4
6	1.11×10^4	8192	4.77×10^6
7	4.45×10^4	32768	1.35×10^9

With $M = 7$, stopping at $T = 60000$ pulls left 75.7% of the runs in round 4 and 24.3% in round 5. Thus, setting a larger M does not mean that the algorithm can actually reach the later rounds within this budget. The parameter choice used in the theory requires still more samples: for example, $M = 6$ and $K(M) = 300$ give $N \approx 2.73 \times 10^8$ and $n_6 = 90990$.

(b) *Delays in admitting new arms.* Arms arriving during round m wait until round $m + 1$. An arm arriving near the beginning of a long round therefore waits for almost its entire length. Proposition 1 bounds this delay under Assumption 2, but it remains unclear why admission should follow the increasingly long elimination rounds, rather than a shorter interval. For example, one could modify the paper's Growing UCB1 baseline to admit waiting arms every fixed H pulls and eliminate arm i when $\mathrm{UCB}_i(t) < \max_{j \in \mathcal{C}_t} \mathrm{LCB}_j(t)$, using each arm's accumulated observations, where $\mathcal{C}_t$ is the set of admitted arms that remain active. A comparison with shorter admission intervals would help establish whether waiting until the end of a round reduces subsequent exploration enough to justify the additional delay.

(c) *The pull count used in the online extension.* In Online UCB-AA, the algorithm is restarted over stages with $M_\ell = 2^\ell$ rounds, and if the interaction ends within the final stage, the

proof completes that stage only for analysis by freezing further arrivals and continuing the auxiliary run to its end. However, Corollary 2 gives a regret bound of order $\sqrt{K(M)T\log T}$ while the proof takes T to be the expected number of pulls in this auxiliary completed-stage run, rather than the expected number of pulls actually made up to the terminal round.

Let T_{act} denote the expected number of pulls actually made and T_{aux} the expected number in the auxiliary completed run. The proof does not establish that they are of comparable order. Since $M_\ell = 2^\ell$, the sampling target at the end of a stage contains a factor 4^{2^ℓ}. If the interaction ends near the beginning of a stage, completing its remaining rounds can therefore require far more pulls than have actually been made, even at an early stage. The authors should clarify whether Corollary 2 is intended as an anytime regret bound in terms of T_{act} and, if so, show how the bound involving T_{aux} implies the claimed rate in terms of T_{act}.

3. Three technical steps in the proofs of the upper bounds need justification.

   (a) *The bound for arms in $G_3(\lambda)$.* On p. 55, Lines 52–56, the proof uses

   $$\frac{8\log(N\widetilde{\Delta}_{m_i}^2)}{\widetilde{\Delta}_{m_i}} + 4\widetilde{\Delta}_{m_i} \le \frac{32\log(N\Delta_i(m_i)^2)}{\Delta_i(m_i)} + 4\Delta_i(m_i).$$

   This step appears to apply a relation at the wrong index. The definition of $\widetilde{m}_i$ gives $\widetilde{\Delta}_{\widetilde{m}_i} < \Delta_i(m_i)/2 \le 2\widetilde{\Delta}_{\widetilde{m}_i}$. This relation holds at $\widetilde{m}_i$, not necessarily at m_i. For $i \in G_3(\lambda)$, $\widetilde{m}_i < m_i$, so $\widetilde{\Delta}_{m_i}$ can be much smaller than $\widetilde{\Delta}_{\widetilde{m}_i}$. In particular, the bound $\Delta_i(m_i) \le 4\widetilde{\Delta}_{\widetilde{m}_i}$ does not justify replacing $1/\widetilde{\Delta}_{m_i}$ by $4/\Delta_i(m_i)$ in the displayed inequality.

   For example, take two initial arms with means $1/2$ and $r_i = 1/2 - 2^{-M}$, and admit an arm of mean 0.9 in round M. With deterministic rewards and $\lambda = 0.1$, for sufficiently large M, we have $t_i = 0$, $m_i = M$, $\widetilde{m}_i = 3 < M$, and $i \in G_3(\lambda)$. The gap is $\Delta_i(m) = 2^{-M}$ for $m < M$ and $\Delta_i(M) = 0.4 + 2^{-M}$. Assumption 1 holds with $\widetilde{C} = 1$. With $K(M) = 3$ and the theoretical parameter choice $N = 30n_M$, both N and n_{M-1} are $\Theta(4^M)$, giving $\widetilde{S}_i = n_{M-1}2^{-M} = \Theta(2^M)$. Thus, this example contradicts the bound

   $$\widetilde{S}_i \le \frac{32\widetilde{C}\log(N\Delta_i(M)^2)}{\Delta_i(M)} + 4\Delta_i(M).$$

   The derivation of Corollary 1, and hence the simplified rate in Theorem 2, therefore needs to be reconsidered.

   (b) *The elimination sequence in Equation (30).* In Equation (30) (p. 50), the event

   $$\bigcap_{k=1}^{s} \{*_k \text{ is eliminated by } *^k\}$$

   involves an elimination sequence determined by the observed rewards. The proof therefore needs to bound the probability that at least one sequence satisfying the constraints in (30) occurs.

   For a fixed sequence,

   $$\Pr\left(\bigcap_{k=1}^{s} \{*_k \text{ is eliminated by } *^k \text{ at phase } m^k\}\right) \le \min_{1\le k\le s} \Pr(*_k \text{ is eliminated by } *^k \text{ at phase } m^k).$$

This inequality is valid, and the individual probabilities can be bounded using the preceding concentration argument. However, the existence of such a sequence is a union over possible sequences. Taking the maximum of the bounds for each fixed sequence does not generally bound this union. Thus, the second inequality in (30) requires further justification. A bound accounting for the data-dependent sequence is needed to justify the error term in Theorem 1.

(c) *The definition of $*(m_i)$ and the comparison in Equation (18).* On p. 8, Lines 8–10, $*(m_i)$ is defined as a member of B_{m_i}, yet the proof considers a case in which this arm was eliminated before that round began (p. 44, Lines 54–57). These statements appear inconsistent and should be reconciled.

If $*(m_i)$ is instead intended to denote the best arm in $A(m_i)$, the comparison in Equation (18) also needs justification. The authors use

$$\Pr\left(\widehat{r}_i(n_{\widetilde{m}_i}) + \sqrt{\frac{\log(N\widetilde{\Delta}^2_{\widetilde{m}_i})}{2n_{\widetilde{m}_i}}} > \widehat{r}_{*(m_i)}(n_{\widetilde{m}_i}) - \sqrt{\frac{\log(N\widetilde{\Delta}^2_{\widetilde{m}_i})}{2n_{\widetilde{m}_i}}}\right),$$

because the eliminated arm's empirical mean has been recorded. However, Algorithm 1 compares each arm's upper confidence bound with the largest lower confidence bound among arms in the current set B^p_m, which excludes arms eliminated in earlier rounds. Keeping an estimate does not itself imply that the upper confidence bound of a surviving new arm must exceed the previously recorded lower confidence bound of the eliminated arm. The event inclusion needs to follow from a comparison the algorithm actually makes.

4. The simulation settings and the practical benefits of the method need to be clarified.

(a) *Comparison with Growing MOSS.* The simulations do not show a clear overall advantage over Growing MOSS. UCB-AA has higher regret and more wasted pulls in Scenario 1, whereas in Scenario 2 it has similar final regret and fewer wasted pulls. The regret curves in Scenario 2 suggest that UCB-AA may compare more favorably over a longer horizon, which would be useful to examine. The authors also emphasize the benefit of maintaining a smaller active set, but its practical value needs further experimental support.

(b) *Differences between the main text and supplement.* The main text and supplement give different settings: $r_0 = 0.6$ versus 0.7 in Scenario 1 (pp. 25 and 92), initial near-optimal means in $[0.7, 0.8]$ versus $[0.89, 0.90]$ in Scenario 2 (pp. 26 and 91–92), and T versus t in the Growing MOSS index (p. 26 and Algorithm 5, p. 94). These differences change the generated data or the comparison method. The specifications used for the figures need to be identified and made consistent.

(c) *The choice of M in the simulations.* The value of M used in the simulations is not reported. In my attempt to reproduce the experiments from the description, I did not fully recover the reported results. The authors should specify the value of M and report the number and lengths of rounds completed within 60000 pulls.

Minor Comments

1. Several definitions and notation choices could be clearer.
   - The relation $\widetilde{\Delta}_m = 2^{-m}$ is not stated explicitly in the main text and must be inferred from the initialization and recursion in Algorithm 1.
   - The simulations introduce μ and related mean parameters without connecting them to r_i (the notation used earlier for an arm's mean reward).
   - On p. 8, Lines 23–25, the description introduces "phases $p = 0, \ldots, m-2$" and "the last two phases $p = m-1, m$" before explaining what one phase consists of.
   - The text refers to completing M rounds, whereas Algorithm 1 runs rounds $0, \ldots, M$, giving $M+1$ rounds.
   - $\widehat{r}_i(n_p)$ in Algorithm 1 (p. 9) is not explicitly defined when first used, and the proof provides no formal definition either. $R^{(i)}(\pi)$ on p. 40 is not standard notation and should be explicitly defined by a formula.
2. The figure presentation could be improved in terms of aesthetics and formatting. For example, labels such as "eps=0.1" could be formatted using mathematical notation, such as $\epsilon = 0.1$.

References

Ghalme, G., Dhamal, S., Jain, S., Gujar, S., and Narahari, Y. (2021). Ballooning multi-armed bandits. *Artificial Intelligence*, 296, 103485.